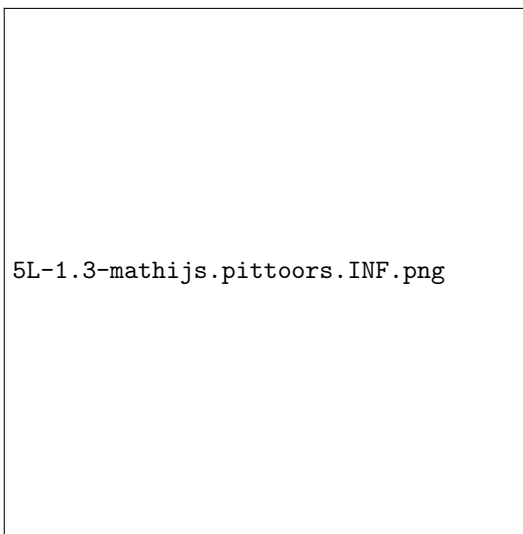


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Studierichting: Informatica
Oefeningenreeks 5L nr. 1.3

$$S = 2\pi \int_0^2 \sqrt{x} \cdot \sqrt{1 + \left(\frac{1}{2\sqrt{x}}\right)^2} dx$$

Tekening



$$= 2\pi \int_0^2 \sqrt{x + \frac{1}{4}} dx$$

Onbepaalde integraal:

$$\text{Neem } t = x + \frac{1}{4} \Rightarrow dt = dx$$

$$\begin{aligned} \int_0^2 \sqrt{x + \frac{1}{4}} dx &= \frac{2}{3} t \sqrt{t} + c = \frac{2}{3} \left(x + \frac{1}{4}\right) \sqrt{x + \frac{1}{4}} + c \\ &= 2\pi \left[\left(x + \frac{1}{4}\right) \sqrt{x + \frac{1}{4}} \right]_0^2 \\ &= \frac{4\pi}{3} \left(\left(\frac{9}{4} \sqrt{\frac{9}{4}}\right) - \left(\frac{1}{4} \sqrt{\frac{1}{4}}\right) \right) \\ &= \frac{13\pi}{3} \end{aligned}$$

References

- [1] <https://www.geogebra.org/m/r4gweqnt>